

A Multi-Agent Architecture for Clinical Document Assessment

System Structure, Agent Roles, and Confidence-Weighted Inter-Agent Agreement

Technical System Documentation

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Research and educational use only. This system is not a medical device and produces no clinical advice.

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1. Abstract

This document describes the structure and operation of a multi-agent system for the assessment of clinical documents. Rather than posing a clinical question to a single language model, the system decomposes the task across a coordinated set of specialised agents: a supervisor that determines which clinical specialties are relevant, a dynamically sized swarm of specialist agents that analyse the source documents independently and in parallel, a judge that consolidates their reviews, and a safety agent that verifies every surviving claim against the original source text.

A distinguishing feature of the architecture is that agent agreement is measured rather than assumed. Each specialist reports a self-assessed confidence, and the system computes a panel-level agreement score by averaging those confidences and discounting the result by their dispersion. This prevents a divided panel from presenting as a confident one, exposes dissenting specialists explicitly, and supplies the downstream judge with a quantitative signal about where the panel diverges.

The system is implemented as a Python service exposing both a request/response API and a streaming event interface, with a browser client that renders each agent as it executes. This document sets out the architecture, the agent roles, the agreement methodology, the event protocol, the validation strategy, and the limitations of the approach.

2. Introduction and Motivation

2.1 Problem statement

A clinical question posed against a patient record is rarely confined to one domain. A single record may simultaneously raise questions of renal function, glycaemic control, drug dosing, and cardiovascular risk. A single general-purpose model answering such a question must hold all of these perspectives at once, and its output offers no way to distinguish a conclusion that several lines of reasoning support from one that rests on a single thread.

Two failure modes motivate the design of this system:

- Ungrounded assertion. A language model may state a clinically plausible fact that does not appear in the supplied documents. Such a claim is indistinguishable in tone from a supported one.
- Undifferentiated confidence. A single model emits one answer with one implicit confidence. There is no way to tell whether that confidence reflects broad support or a narrow, contestable inference.

2.2 Design response

The architecture addresses the first failure mode with a dedicated verification stage that treats the source documents as the sole ground truth, and the second by replacing a single reasoner with a panel whose members answer independently and whose degree of agreement is computed explicitly. Neither mechanism makes the system clinically safe; both make its output easier to interrogate.

3. System Architecture

3.1 Processing pipeline

Execution proceeds through four sequential stages. Stages 1, 3, and 4 each invoke a single agent; stage 2 invokes a variable number of agents concurrently. Agreement scoring is a deterministic computation performed between stages 2 and 3 and involves no model call.

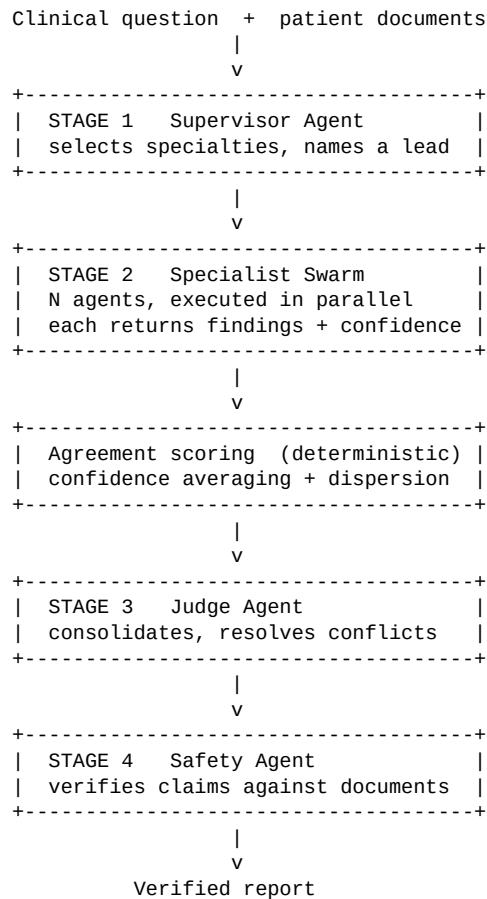


Figure 1. Four-stage processing pipeline with deterministic agreement scoring.

3.2 Agent roles

Every agent derives from a common base class that handles model invocation, optional token-level streaming, structured JSON parsing, and prompt assembly. Agents differ in their system prompt, their output schema, and the model assigned to them.

Agent	Stage	Responsibility	Structured output
Supervisor	1	Reads the question and documents; determines which clinical specialties are required and designates one as lead.	specialties[], lead_specialist, rationale
Specialist	2	Instantiated once per specialty. Analyses the documents from a single domain perspective, quoting supporting evidence.	specialty, findings, recommendation, evidence_quotes[], confidence
Judge	3	Consolidates all specialist reviews into one coherent report, resolving conflicts between them.	final_summary, key_findings_by_specialty, consolidated_recommendations[]
Safety	4	Verifies each claim in the judge report against the original documents and flags anything unsupported.	verified_findings[], flagged_issues[], final_summary, overall_confidence, is_safe

Table 1. Agent responsibilities and output schemas.

3.3 Dynamic swarm construction

The number of specialist agents is not fixed. It is determined at run time by the supervisor and bounded by a configurable ceiling (MAX_SPECIALISTS, default 10) to prevent unbounded cost on an ambiguous input. A record raising a single domain question produces one specialist; a complex multi-morbidity record may produce several. Specialists are dispatched concurrently, so wall-clock latency is governed by the slowest agent rather than their sum.

3.4 Externalised prompts

Agent instructions are not embedded in source code. Each role has a directory under skills/ containing a Markdown instruction file and an optional references/ folder of supporting material. Both are loaded into the system prompt when the agent is constructed. Agent behaviour can therefore be revised without modifying Python, which keeps prompt engineering separable from application logic and legible to non-programmers.

4. Confidence-Weighted Inter-Agent Agreement

This section describes the mechanism by which the system quantifies agreement across the specialist panel. It is the component that distinguishes this architecture from a conventional sequential agent pipeline.

4.1 Why a plain mean is insufficient

Each specialist reports a self-assessed confidence c in the interval $[0, 1]$. The naive aggregate is the arithmetic mean. This is inadequate because the mean is insensitive to the distribution that produced it. Consider two panels of two specialists each:

Panel	Reported confidences	Arithmetic mean	Interpretation
A	0.50 and 0.50	0.50	Both specialists are moderately confident. The panel concurs.
B	0.10 and 0.90	0.50	One specialist is nearly certain, the other nearly certain of the opposite. The panel is split.

Table 2. Two panels with identical means but incompatible interpretations.

The two panels are epistemically very different, yet the mean reports them identically. Presenting panel B as a 50% result conceals a genuine disagreement that a reader would want to see. The aggregation must therefore be sensitive to dispersion as well as to central tendency.

4.2 Definition

Let $C = \{c_1, \dots, c_n\}$ be the confidences of the n participating specialists. The agreement score A is defined as the mean confidence discounted by a normalised measure of dispersion:

$$\begin{aligned}\text{mean} &= (1/n) * \text{SUM } c_i \\ \text{sigma} &= \text{population standard deviation of } C \\ \text{dispersion} &= \min(\text{sigma} / 0.5 , 1) \\ A &= \text{mean} * (1 - \text{dispersion})\end{aligned}$$

The normalising constant 0.5 is the maximum standard deviation attainable by values bounded to $[0, 1]$, which occurs when half the panel reports 0 and half reports 1. Dividing by it maps sigma onto $[0, 1]$, so dispersion is a proportion rather than an unbounded quantity, and a maximally divided panel receives an agreement score of zero irrespective of how confident its individual members were.

The population standard deviation is used in preference to the sample standard deviation because the panel is the entire population of interest, not a sample drawn from a larger one.

4.3 Behaviour

Applying the definition to the two motivating panels, and to two further cases:

Confidences	Mean	sigma	Dispersion	Agreement A	Level
0.50, 0.50	0.500	0.000	0.00	0.500	moderate
0.10, 0.90	0.500	0.400	0.80	0.100	none
0.90, 0.85	0.875	0.025	0.05	0.831	strong
0.70 (single agent)	0.700	0.000	0.00	0.700	moderate

Table 3. Agreement scores for representative panels.

The two panels that share a mean of 0.500 are now separated by a factor of five. A single-agent panel has no dispersion and therefore scores its own confidence unmodified, which is the correct degenerate case: with no second opinion there is nothing to agree or disagree with.

4.4 Abstention

A specialist that finds nothing relevant in its domain is instructed to report a confidence of zero. Such a response is treated as an abstention and excluded from both the mean and the dispersion. This distinction is material: a dermatologist finding no dermatological content in a cardiology record is not disagreeing with the cardiologist, and admitting that zero into the mean would depress the panel score for a reason unrelated to consensus. Abstentions are counted and reported separately so that they remain visible.

4.5 Outlier identification

A specialist whose confidence lies more than one standard deviation from the panel mean is labelled an outlier, subject to a minimum absolute deviation of 0.15. The floor is necessary because standard deviation is scale-relative: in a tightly clustered panel reporting 0.95, 0.85 and 0.85, the standard deviation is approximately 0.047, and the specialist at 0.95 sits more than one such deviation from the mean despite differing from its colleagues by a clinically negligible margin. Without the floor the system would label ordinary variation as dissent. The floor was introduced after this behaviour was observed in a live run of the system.

4.6 Interpretation bands

The continuous score is additionally reported as a qualitative band, for presentation purposes only; the underlying score is always retained.

Band	Range of A	Reading
strong	$A \geq 0.75$	Specialists are confident and closely aligned.

Band	Range of A	Reading
moderate	$0.50 \leq A < 0.75$	Broad alignment, or alignment at moderate confidence.
weak	$0.25 \leq A < 0.50$	Meaningful divergence or generally low confidence.
none	$A < 0.25$	The panel is divided, or no specialist reported confidence.

Table 4. Qualitative agreement bands.

4.7 Use of the score

The agreement report is consumed at three points. It is emitted to the client as a discrete event so the interface can display it as soon as the swarm completes; it is injected into the judge prompt, with an instruction to state explicitly where the panel diverges rather than smoothing the disagreement away; and it is returned in the final API response for downstream analysis.

5. Execution Model and Event Protocol

The system exposes two interfaces over the same pipeline. A request/response endpoint runs the pipeline to completion and returns the aggregate result. A streaming endpoint emits a typed event at every state transition, allowing a client to render the pipeline as it executes rather than after it finishes.

Event	Emitted when
triage_thinking / triage_done	The supervisor begins and completes specialty selection.
specialists_spawned	The swarm has been constructed; carries the specialty list and lead.
specialist_thinking / specialist_done	Once per specialist, on dispatch and on completion.
consensus_done	Agreement scoring has been computed over the completed swarm.
judge_thinking / judge_done	The judge begins and completes consolidation.
safety_thinking / safety_done	The safety agent begins and completes verification.
agent_stream	Per token, for any agent generating output.
pipeline_complete	Terminal event carrying the full aggregate result.

Table 5. Streaming event types.

Because agreement scoring is deterministic and requires no model call, `consensus_done` is emitted immediately upon completion of the final specialist, with no perceptible delay between the swarm finishing and the agreement panel appearing.

6. Implementation

6.1 Technology

Layer	Technology	Role
Service	Python 3.11+, FastAPI, Uvicorn	HTTP and WebSocket interfaces, request validation.
Agents	OpenAI API, asyncio	Model invocation; concurrent dispatch of the swarm.
Contracts	Pydantic	Typed schemas for every agent output and API payload.
Configuration	pydantic-settings	Environment-driven settings with typed defaults.
Client	React 19, Vite	Live rendering of pipeline state.
Testing	pytest, pytest-asyncio	Unit and integration tests against mocked model responses.

Table 6. Implementation stack.

6.2 Repository structure

agents/	
base.py	shared model client, streaming, JSON parsing
supervisor.py	stage 1
specialist.py	stage 2, plus swarm factory
judge.py	stage 3
safety.py	stage 4
orchestrator/	
pipeline.py	four-stage coordinator
consensus.py	agreement scoring (section 4)
events.py	typed streaming events
api/	
main.py	REST and WebSocket endpoints
schemas.py	request and response contracts
config/	
settings.py	environment-driven configuration
skills/	externalised agent prompts (section 3.4)
frontend/	React client
tests/	test suite

6.3 Configuration

All operational parameters are supplied through environment variables and read into a typed settings object. Model selection is per-role, so the supervisor and safety agents can be assigned different models from the specialists. Transport verification, request timeout, swarm ceiling, and permitted client origins are likewise configurable.

7. Validation

The system is exercised at three levels. Agent-level tests confirm that each role parses well-formed model output correctly and degrades gracefully on malformed output. Pipeline-level tests run all four stages against scripted model responses and assert the aggregate result, including the computed agreement figures. Interface-level tests exercise the API endpoints and their schemas.

The agreement component is tested independently against the cases in section 4, together with adversarial inputs: absent confidence fields, non-numeric values, values outside $[0, 1]$, panels in which every member abstains, and the empty panel. In each case the scoring must return a well-formed report rather than raise. At the time of writing the suite comprises twenty tests, all passing.

Automated tests are supplemented by end-to-end execution against the live model API, since mocked responses cannot demonstrate that the deployed system functions. The outlier floor described in section 4.5 was added as a direct result of such a run.

8. Limitations and Ethical Considerations

8.1 Limitations of the method

- Self-reported confidence is not calibrated. The agreement score aggregates what agents assert about their own certainty, which is a stated disposition rather than a measured probability. A uniformly overconfident panel yields a high agreement score.
- Agreement is not correctness. Independently reasoning agents drawing on a shared underlying model may share its errors, and concordance among them offers no protection against a systematic mistake. A high score indicates consistency, not validity.
- Confidence is scalar. The score measures dispersion in stated certainty, not semantic disagreement in the findings themselves. Two specialists may report identical confidence while advancing incompatible recommendations.
- Verification is bounded by the documents. The safety agent can only establish that a claim is or is not supported by the supplied text. It cannot detect a clinically incorrect statement that the documents happen to support, nor an omission.

8.2 Intended use

The system is a research and educational artefact. It is not a medical device, has not been clinically validated, and must not be used to inform patient care. Any material processed by it is transmitted to a third-party model provider, which precludes the use of identifiable patient data absent an appropriate legal and technical basis. The safety stage reduces the incidence of ungrounded assertion; it does not render the output reliable, and no stage of the pipeline substitutes for review by a qualified clinician.

9. Conclusion

The architecture described here decomposes clinical document assessment across four stages and a dynamically sized panel of specialist agents, and makes two properties of the result explicit that a single-model approach leaves implicit: whether each claim is grounded in the source documents, and how far the panel agreed in producing it.

The agreement mechanism is deliberately simple. It requires no additional model call, is fully deterministic and therefore reproducible, and is legible enough that a reader can reconstruct the score by hand from the reported confidences. Its limitations follow from that simplicity: it aggregates stated rather than calibrated confidence, and it measures consistency rather than correctness. Within those bounds it supplies something a single opinion cannot, namely an explicit and inspectable signal of when the constituent analyses did not converge.